

MASHHAD INTL, Iran, Islamic Republic of

WMO: 407450

Lat: 36.2366N

Lon: 59.6317E

Elev: 999

StdP: 89.89

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-5.9	-15.1	1.1	3.2	-12.2	1.5	5.6	9.4	5.5	8.0	6.2	1.4	150	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	37.8	17.5	36.2	17.3	35.1	16.9	21.2	33.2	20.0	32.2	19.0	31.1	5.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.7	28.4	15.2	12.2	26.3	14.1	11.4	25.1	66.6	33.6	62.0	32.3	58.1	31.1	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	6.8	-10.7	40.1	4.0	1.2	-13.6	41.0	-15.9	41.7	-18.2	42.4	-21.1	43.3		
(5)			-11.2	21.5	4.0	1.9	-14.0	22.9	-16.4	24.0	-18.6	25.1	-21.5	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	3.4	5.2	10.1	15.6	21.5	26.3	28.3	26.6	22.1	16.1	9.4	5.1		
	DBStd	9.56	4.78	4.92	4.98	4.82	3.59	2.63	2.19	2.33	3.14	4.14	4.41	3.94		
	HDD10.0	646	208	143	61	11	1	0	0	0	0	5	61	158		
	HDD18.3	1983	463	367	257	105	13	1	0	0	7	92	268	412		
	CDD10.0	2788	3	9	64	179	356	488	567	516	364	195	44	4		
	CDD18.3	1084	0	0	2	23	109	239	308	257	121	23	1	0		
	CDH23.3	13564	0	1	35	277	1158	2980	4158	3286	1341	311	19	0		
(13)	CDH26.7	6942	0	0	5	79	468	1579	2367	1777	573	89	3	0		
(14)	Wind	WSAvg	2.8	2.1	2.7	2.9	3.0	3.4	3.6	3.4	2.8	2.4	2.0	1.9		
(15) Precipitation	PrecAvg	218	24	33	52	40	28	3	2	1	2	10	18	23		
	PrecMax	309	78	130	116	130	116	47	17	6	26	59	62	78		
	PrecMin	117	0	3	4	4	0	0	0	0	0	0	0	0		
	PrecStd	61	18	27	28	27	25	8	4	2	5	14	17	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	20.8	26.8	32.0	35.8	38.9	39.8	38.9	36.0	32.1	26.0	19.2		
		MCWB	7.8	9.7	12.7	15.4	16.4	17.6	18.0	17.7	15.9	15.2	12.8	8.4		
	2%	DB	15.2	17.8	23.8	29.0	33.1	36.9	38.1	36.9	33.9	29.1	22.0	16.1		
		MCWB	6.8	8.8	11.3	15.2	16.0	17.0	17.6	17.5	15.7	14.2	10.7	7.2		
	5%	DB	13.0	15.2	21.1	26.7	31.2	35.2	36.9	35.6	32.0	26.8	19.2	13.9		
		MCWB	6.0	7.4	10.6	14.2	15.7	16.4	17.6	17.4	15.3	13.4	9.8	6.6		
	10%	DB	10.8	13.1	18.9	24.1	29.2	33.9	35.2	34.1	30.1	24.4	17.0	11.8		
		MCWB	4.9	6.7	9.9	13.2	15.1	16.3	17.4	16.7	14.9	12.5	8.9	5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	11.3	14.6	17.6	19.7	20.8	23.5	23.0	20.5	16.6	13.7	9.6		
		MCDB	14.8	16.7	22.4	27.1	29.4	34.0	36.1	33.3	32.1	27.6	23.1	16.6		
	2%	WB	7.7	9.7	12.8	16.2	17.9	19.6	21.6	21.1	18.8	15.1	12.1	8.4		
		MCDB	13.8	15.4	20.0	25.4	28.1	31.9	33.6	33.0	29.6	26.1	19.1	14.4		
	5%	WB	6.6	8.4	11.6	15.3	17.0	18.6	20.1	19.7	17.1	14.1	11.0	7.2		
		MCDB	11.8	13.7	18.7	23.6	26.9	30.7	32.6	32.1	27.8	24.5	17.0	12.4		
	10%	WB	5.3	7.3	10.6	14.2	16.2	17.7	19.0	18.2	15.7	13.2	9.7	6.2		
		MCDB	9.9	12.1	17.4	22.1	26.1	30.0	31.5	30.7	26.9	22.6	15.6	10.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.4	11.1	11.8	13.1	13.9	13.5	14.5	14.8	13.9	11.5	10.2		
		MCDBR	14.1	14.2	15.3	15.8	15.8	15.9	15.1	16.0	17.0	16.9	15.2	14.1		
		MCWBR	7.3	6.9	6.2	6.0	5.5	5.3	5.2	6.2	6.7	7.2	6.8	7.2		
		MCDBR	12.4	12.3	13.4	13.5	13.9	13.9	13.2	14.6	15.0	14.8	12.9	12.2		
	5% WB	MCWBR	6.9	6.6	6.1	5.8	5.4	6.0	6.5	7.8	7.4	6.7	6.2	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.335	0.378	0.454	0.498	0.516	0.494	0.486	0.440	0.404	0.423	0.381	0.333		
	taud		2.278	2.139	1.938	1.876	1.858	1.907	1.934	2.045	2.118	2.045	2.148	2.292		
	Ebn at Noon		844	842	805	793	784	799	803	833	844	779	775	819		
	Edh at Noon		101	131	176	197	204	194	188	164	144	142	114	94		
(44) All-Sky Solar Radiation	RadAvg		2.49	3.28	4.37	5.70	6.92	8.03	7.94	7.43	6.18	4.44	2.98	2.33		
	RadStd		0.22	0.40	0.34	0.53	0.36	0.27	0.18	0.22	0.17	0.23	0.25	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.43	N/A	-1.80	+0.34	-1.81	-1.59	N/A	N/A	N/A	+89
(47)	Regional (7 neighbors)	N/A	N/A	N/A	+0.50	-0.74	-0.72	N/A	N/A	N/A	+88

Nomenclature: See separate page